

RETHINKING CLASSIFIER-FREE GUIDANCE IN ON-POLICY DIFFUSION DISTILLATION

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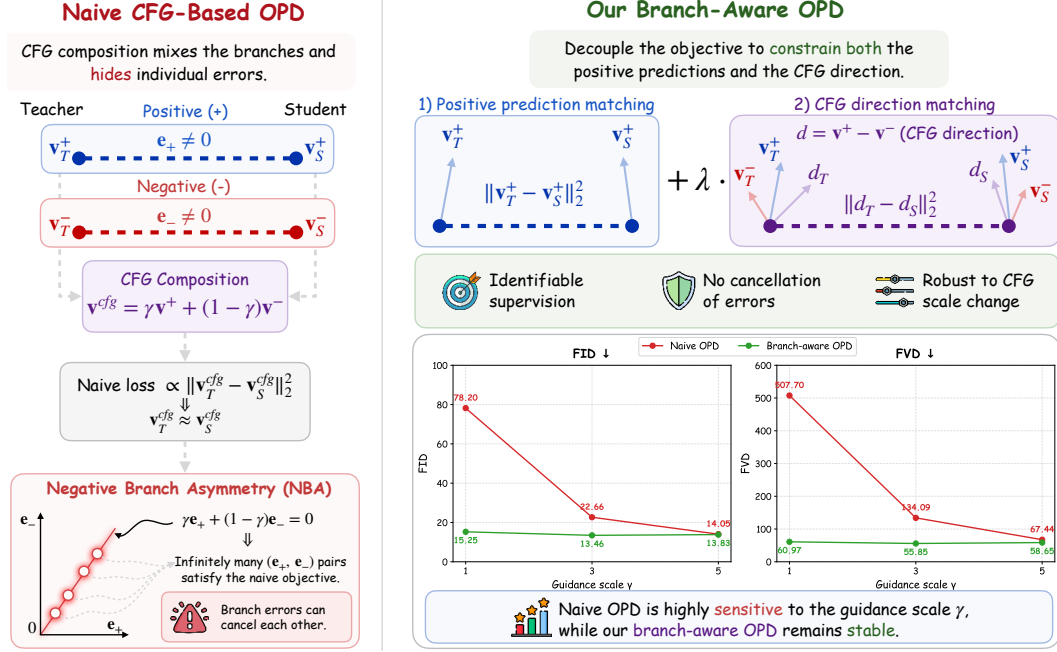


Figure 1: **Naive CFG-based OPD suffers from branch ambiguity.** Matching only the CFG-composed prediction allows positive and negative branch errors to cancel, creating infinitely many degenerate solutions. Our branch-aware OPD resolves this ambiguity by separately constraining branch predictions and CFG directions, leading to identifiable supervision and robust distillation across guidance scales.

ABSTRACT

On-policy distillation (OPD) has recently emerged as a promising paradigm for adapting diffusion models by querying a teacher along trajectories generated by the current student. However, its formulation under classifier-free guidance (CFG), a fundamental component of modern diffusion systems, remains poorly understood. Existing OPD methods naturally extend velocity matching to the CFG-composed prediction, matching teacher and student guided velocities directly. We show that this objective is valid in standard text-to-image settings, where teacher and student remain aligned on the negative branch, but can fail in more general settings with asymmetric teacher–student conditioning. In such cases, CFG composition couples positive- and negative- branch discrepancies, allowing branch-level errors to cancel while producing identical guided supervision signals. We term this phenomenon Negative Branch Asymmetry (NBA). To address NBA, we introduce Positive–Direction Matching (PDM), a branch-aware OPD objective that separately constrains the positive prediction and the CFG conditional direction. Experiments on FLUX-based visual knowledge distillation and dense-to-sparse video control demonstrate that naive guided matching suffers from CFG-scale sensitivity under NBA, while branch-aware objectives enable more robust and effective knowledge transfer.

1 INTRODUCTION

On-policy distillation (OPD) has recently emerged as a promising paradigm for adapting diffusion models without relying on fixed teacher-generated trajectories or paired offline datasets (Jiang et al., 2026; Fang et al., 2026; Li et al., 2026b). Unlike offline distillation, diffusion OPD first lets the student generate a denoising trajectory using its current policy. At each state visited by this trajectory, the teacher and student are evaluated on the same student-visited noisy state, and the student is updated to match the teacher’s local velocity field. This combination of on-policy state coverage and dense per-step supervision makes OPD an appealing alternative to offline distillation and scalar-reward policy optimization (Agarwal et al., 2024; Zhong et al., 2026).

A key question for applying OPD to modern diffusion systems is whether the derived velocity-matching objective remains valid for the guided denoising process used in practice. Unlike the generic velocity field considered in the OPD formulation, modern diffusion models (Labs, 2025; Esser et al., 2024; Wan et al., 2025) are commonly deployed with classifier-free guidance (CFG) (Ho & Salimans, 2022), where the actual denoising velocity is constructed by combining positive- and negative- branch predictions. Despite its widespread use, how CFG changes the OPD supervision target remains poorly understood.

For CFG-enabled diffusion models, existing OPD methods (Fang et al., 2026; Li et al., 2026b) naturally apply the velocity-matching objective to the guided prediction used during denoising:

$$\mathcal{L}_{\text{naive}} \propto \|\mathbf{v}_{\text{cfg}}^T - \mathbf{v}_{\text{cfg}}^S\|_2^2,$$

where $\mathbf{v}_{\text{cfg}}$ denotes the CFG-composed velocity. This effectively treats the CFG-composed velocity field as the supervision target for OPD.

We show that in standard text-to-image distillation settings studied in prior work (Fang et al., 2026; Li et al., 2026b), where teacher and student share identical conditioning interfaces and are instantiated from the same base diffusion model, the CFG-based objective reduces to a scaled positive-branch velocity matching loss. In this regime, the negative branch discrepancy is negligible, and the naive objective reduces to a scaled form of the original OPD velocity matching objective, and explains why the method performs well in standard text-to-image settings.

However, in more general OPD settings where the teacher and student differ in architecture, capacity, or access to conditioning information, this equivalence breaks down. In such cases, the CFG-composed velocity couples discrepancies from both positive and negative branches, causing the naive objective to deviate from true velocity alignment and leading to degraded performance when applied across different CFG scales during inference.

We refer to this phenomenon as *Negative Branch Asymmetry* (NBA), where asymmetry in the negative branch between teacher and student causes the CFG-based distillation objective to only enforce alignment of the final guided velocity, while failing to separately constrain the underlying positive- and negative- branch prediction errors. This leads to a loss of identifiability at the branch level, where different combinations of positive- and negative- branch discrepancies can produce the same guided supervision signal.

To resolve this ambiguity, we introduce branch-aware OPD objectives that constrain teacher–student discrepancies before CFG composition. Our primary formulation, Positive-Direction Matching (PDM), aligns the positive prediction while separately preserving the conditional direction encoded by CFG. Rather than supervising the student only after its positive and negative predictions have been merged, our objective prevents cross-branch error cancellation and removes the resulting ambiguity in supervision. It thereby retains the useful conditional signal of CFG while providing a well-constrained objective for diffusion OPD.

We investigate NBA in two substantially different settings. First, we perform D-OPSD-based (Jiang et al., 2026) visual knowledge distillation to a multi-step FLUX (Labs, 2025) base model and perform distillation from a reference-conditioned teacher to a text-only student. Second, we study dense-to-sparse video control distillation, where the teacher observes dense pose guidance while the student receives only sparse controls. In both settings, the privileged visual condition changes not only the positive teacher–student pair but also their negative branches, placing them outside the benign regime assumed by naive guided matching. Across image and video experiments, the naive objective exhibits the predicted failure, whereas our PDM objective enables stable and effective

knowledge transfer. These results demonstrate that NBA is not tied to a particular modality or conditioning mechanism, but can arise whenever architectural, capacity, or conditioning asymmetries between the teacher and student produce a non-negligible discrepancy on the negative branch.

Our contributions are summarized as follows:

- We identify a hidden negative-branch alignment condition underlying naive CFG-based diffusion OPD, and distinguish the benign configurations in which this condition holds from more general settings in which it is violated.
- We formalize *Negative Branch Asymmetry* as a teacher–student discrepancy on the negative branch induced by architectural, capacity, or conditioning asymmetries, and show that it creates infinitely many degenerate solutions through cross-branch error cancellation under naive guided matching.
- We introduce Positive–Direction Matching, a branch-aware OPD objective that removes cross-branch cancellation by supervising the positive prediction and CFG conditional direction. We additionally study independent branch matching as a direct branch-aware baseline.

2 RELATED WORK

2.1 DIFFUSION MODEL DISTILLATION

Although our goal is not few-step generation, prior diffusion distillation provides the closest context for understanding how teacher supervision and classifier-free guidance are typically transferred to a student. Most existing methods focus on reducing the number of denoising steps required at inference. Progressive Distillation recursively compresses multiple teacher steps into a single student step (Salimans & Ho, 2022). Consistency-based methods, including Consistency Models (Song et al., 2023; Kim et al., 2024) and Latent Consistency Models (Luo et al., 2023), train a model to produce consistent outputs across states along the same probability-flow trajectory, enabling one- or few-step generation. Distribution Matching Distillation (DMD) matches the student distribution to the teacher through their score functions (Yin et al., 2024b), while other approaches further combine diffusion supervision with adversarial training (Sauer et al., 2024; Yin et al., 2024a). Many of these methods also distill classifier-free guidance into the student. Rather than retaining separate positive and negative branches at inference, they use the CFG-composed teacher prediction as the distillation target, thereby baking the effect of guidance into a single student prediction (Luo et al., 2023; Yin et al., 2024b; Wu et al., 2026; Huang et al., 2025). This design improves sampling efficiency by avoiding two model evaluations for CFG and, more broadly, transfers the strongly guided teacher distribution into a compact few-step generator.

Our setting differs in both objective and purpose. We do not primarily seek to compress a multi-step guided teacher into a CFG-free few-step student. Instead, we study diffusion OPD in which the teacher and student each construct their own guided prediction from positive and negative branches. Under this formulation, matching the final guided predictions may fail to constrain the underlying positive branch when the two models are asymmetric on the negative branch.

2.2 ON-POLICY DISTILLATION FOR DIFFUSION MODELS

Recent work extends on-policy distillation to diffusion and flow models by querying the teacher on states sampled from the current student. D-OPSD performs on-policy self-distillation using privileged context, allowing a student to learn from a stronger teacher evaluated along the student’s own denoising trajectory (Jiang et al., 2026). Flow-OPD distills task-specialized flow-matching teachers into a unified student through on-policy sampling and dense per-step supervision (Fang et al., 2026). DiffusionOPD formulates this process through transition-level KL divergence and shows that it reduces to prediction matching under standard diffusion transitions (Li et al., 2026b).

Despite differences in derivation and optimization, these methods share the same central principle: the student generates the trajectory, while the teacher provides dense local supervision at student-visited states. Their studied configurations, however, either avoid explicit CFG during distillation or keep the teacher and student sufficiently aligned on the negative branch. They therefore do not examine the more general case where differences in architecture, model capacity, or conditioning inputs produce a non-negligible negative-branch discrepancy.



Figure 2: Qualitative manifestation of Negative Branch Asymmetry (NBA) under guidance-scale shifts. We compare PDM with naive CFG-based OPD on (a) visual knowledge distillation ($\gamma_{\text{train}} = 3$) and (b) dense-to-sparse video control ($\gamma_{\text{train}} = 5$). Naive OPD degrades when the inference guidance scale differs from the training configuration, exposing the branch-error cancellation permitted by guided-only matching. PDM remains more stable by separately matching the positive prediction and CFG conditional direction.

Our work identifies this overlooked regime as NBA. We show that, under such asymmetry, directly matching the CFG-composed teacher and student predictions admits cross-branch cancellation and fails to uniquely constrain the positive branch. We further introduce branch-aware OPD objectives, with Positive-Direction Matching as the primary formulation, to resolve this ambiguity under negative-branch mismatch.

3 PRELIMINARIES

3.1 ON-POLICY DISTILLATION FOR DIFFUSION MODELS

Diffusion on-policy distillation (OPD) trains a student model using states sampled from its own evolving generation process (Jiang et al., 2026; Fang et al., 2026; Li et al., 2026b). Let

$$\mathbf{x}_t^S \sim p_\theta \quad (1)$$

denote a noisy state visited along a trajectory generated by the current student. At this state, the teacher and student define local denoising transitions conditioned on $\mathbf{c}_T$ and $\mathbf{c}_S$, respectively. The teacher may have access to privileged information unavailable or only partially available to the student, and therefore $\mathbf{c}_T$ and $\mathbf{c}_S$ need not be identical.

Existing diffusion OPD (Jiang et al., 2026; Fang et al., 2026; Li et al., 2026b) formulations minimize the discrepancy between the teacher and student transitions at student-visited states. Although they differ in their derivations and optimization procedures, their transition-level KL objectives reduce, up to a timestep-dependent weighting factor, to matching the corresponding model predictions:

$$\mathcal{L}_{\text{OPD}} = \mathbb{E}_{\mathbf{x}_t^S \sim p_\theta, t} \left[w(t) \left\| \mathbf{v}_T(\mathbf{x}_t^S, t, \mathbf{c}_T) - \mathbf{v}_S(\mathbf{x}_t^S, t, \mathbf{c}_S) \right\|_2^2 \right], \quad (2)$$

where $\mathbf{v}_T$ and $\mathbf{v}_S$ denote the teacher and student velocity predictions, and $w(t)$ absorbs the parameterization- and solver-dependent weighting. We use velocity prediction throughout the paper, while the same formulation applies to equivalent noise, score, or flow parameterizations.

3.2 DIFFUSION OPD WITH CLASSIFIER-FREE GUIDANCE

We next consider diffusion models that use classifier-free guidance (CFG), where the velocity used for denoising is

$$\tilde{\mathbf{v}} = \gamma \mathbf{v}^+ + (1 - \gamma) \mathbf{v}^-. \quad (3)$$

Define the positive- and negative-branch discrepancies as

$$\mathbf{e}_+ = \mathbf{v}_T^+ - \mathbf{v}_S^+, \quad \mathbf{e}_- = \mathbf{v}_T^- - \mathbf{v}_S^-. \quad (4)$$

Substituting the branch discrepancies into the CFG objective gives

$$\mathcal{L}_{\text{naive}} = \mathbb{E}_{\mathbf{x}_t^S \sim p_{\theta}, t} [w(t) \|\gamma \mathbf{e}_+ + (1 - \gamma) \mathbf{e}_-\|_2^2]. \quad (5)$$

Existing OPD settings (Fang et al., 2026; Li et al., 2026b) are primarily studied in text-to-image distillation, where teacher and student are instantiated from the same base diffusion model (e.g., SD3.5 (Esser et al., 2024)) and share identical null-text conditioning in the negative branch. In this regime, the negative-branch discrepancy $\mathbf{e}_-$ is typically small, causing CFG-based supervision to approximately reduce to positive-branch matching.

4 METHOD

4.1 NEGATIVE BRANCH ASYMMETRY

In general diffusion OPD settings, the teacher and student may exhibit asymmetries in model architecture, capacity, or conditioning inputs. In such cases, Eq. 5 can no longer be approximated as positive-branch matching. Instead, zero guided loss only implies

$$\mathbf{e}_+ = \frac{\gamma - 1}{\gamma} \mathbf{e}_-, \quad (6)$$

which admits infinitely many non-zero solutions in the joint error space.

This implies that supervision on the CFG-composed direction is insufficient to uniquely constrain the positive-branch error, since errors in the negative branch can be arbitrarily traded off against positive-branch errors without affecting the objective value. We therefore refer to this phenomenon as **Negative Branch Asymmetry (NBA)**.

4.2 BRANCH-AWARE OPD

Based on this observation, we introduce Positive–Direction Matching (PDM), our primary branch-aware objective. PDM supervises the positive prediction together with the CFG conditional direction, thereby removing the cancellation ambiguity of guided-only matching. To contextualize this design, we additionally consider Independent Branch Matching (IBM), a straightforward baseline that directly matches the positive and negative branches separately.

For clarity, we use ℓ to denote the pointwise distillation loss evaluated at a single student-visited state $\mathbf{x}_t^S$ and timestep t . The corresponding OPD training objective is obtained by averaging this loss over the student-induced state distribution and timesteps:

$$\mathcal{L}_{\text{OPD}}[\ell] = \mathbb{E}_{\mathbf{x}_t^S \sim p_{\theta}, t} [w(t) \ell(\mathbf{x}_t^S, t)]. \quad (7)$$

All branch-aware objectives below specify different choices of the pointwise loss ℓ , while sharing the same on-policy sampling and timestep weighting in Eq. 7.

Positive–Direction Matching (PDM). Our primary branch-aware objective is Positive–Direction Matching (PDM), which supervises the positive prediction together with the CFG conditional direction.

For a model $M \in \{T, S\}$, we define its conditional direction as

$$\mathbf{d}_M = \mathbf{v}_M^+ - \mathbf{v}_M^-. \quad (8)$$

The pair $(\mathbf{v}_M^+, \mathbf{d}_M)$ uniquely determines the underlying positive and negative predictions, and therefore the CFG-composed prediction at any guidance scale. The corresponding teacher–student direction discrepancy is

$$\mathbf{d}_T - \mathbf{d}_S = \mathbf{e}_+ - \mathbf{e}_-. \quad (9)$$

We define the PDM loss as

$$\ell_{\text{PDM}} = \|\mathbf{e}_+\|_2^2 + \lambda \|\mathbf{d}_T - \mathbf{d}_S\|_2^2 = \|\mathbf{e}_+\|_2^2 + \lambda \|\mathbf{e}_+ - \mathbf{e}_-\|_2^2, \quad (10)$$

where $\lambda > 0$ controls the strength of conditional-direction matching.

The first term anchors the positive prediction, while the second preserves the CFG conditional direction. For $\lambda > 0$, zero PDM loss requires $\mathbf{e}_+ = \mathbf{e}_- = 0$, thereby eliminating the cancellation ambiguity of naive guided matching.

Alternative baseline: Independent Branch Matching (IBM). For comparison, we consider Independent Branch Matching (IBM), a direct branch-aware baseline that matches the positive and negative predictions separately:

$$\ell_{\text{IBM}} = \gamma^2 \|\mathbf{e}_+\|_2^2 + (\gamma - 1)^2 \|\mathbf{e}_-\|_2^2. \quad (11)$$

For the standard guided regime $\gamma > 1$, IBM has the same unique branch-level zero-loss solution as PDM. The two objectives remove the same cancellation ambiguity but parameterize and weight the supervision differently; we compare their behavior empirically in Table 1.

Efficient partial-trajectory supervision. Applying OPD supervision to every state along a full denoising trajectory requires repeated teacher evaluations and substantially increases training cost. We empirically find that such full-trajectory supervision is unnecessary: using only a short prefix of student-visited states can preserve distillation quality while greatly improving training efficiency. Motivated by this observation, we introduce partial-trajectory supervision.

Let $\{\mathbf{x}_{t_k}^S\}_{k=0}^{N-1}$ denote the states visited by the student in rollout order. We accumulate the distillation objective over only the first K states:

$$\mathcal{L}_{\text{OPD}}^{(K)} = \mathbb{E}_{\{\mathbf{x}_{t_k}^S\}_{k=0}^{N-1} \sim p_\theta} \left[\frac{1}{K} \sum_{k=0}^{K-1} w(t_k) \ell(\mathbf{x}_{t_k}^S, t_k) \right], \quad (12)$$

where ℓ can be any pointwise OPD objective defined above. The student still completes the standard denoising rollout, but teacher supervision is computed only for these K states. The hyperparameter K therefore controls the trade-off between supervision coverage and training efficiency, with $K = N$ recovering full-trajectory supervision. As shown in Section 5.3.2, a moderate horizon substantially reduces training cost while maintaining strong distillation performance.

4.3 TASK INSTANTIATIONS

4.3.1 VISUAL KNOWLEDGE DISTILLATION

We consider D-OPSD-based (Jiang et al., 2026) visual knowledge distillation using a multi-step FLUX (Labs, 2025) model with CFG. The teacher observes a reference image $\mathbf{r}$ containing visual knowledge that is not fully specified by the text prompt, while the student is conditioned only on the text prompt $\mathbf{y}$. The reference-supplied knowledge may correspond to a transferable rendering style and the visual identity of a particular concept (Wang et al., 2026). Their positive conditions are therefore

$$\mathbf{c}_T^+ = (\mathbf{y}, \mathbf{r}), \quad \mathbf{c}_S^+ = \mathbf{y}. \quad (13)$$

Importantly, the reference image remains available to the teacher when constructing its negative prediction:

$$\mathbf{c}_T^- = (\emptyset, \mathbf{r}), \quad \mathbf{c}_S^- = \emptyset, \quad (14)$$

where $\emptyset$ denotes the null-text condition. Consequently, the teacher and student are asymmetric not only on the positive branch but also on the negative branch, making this setting a direct instance of NBA.

4.3.2 DENSE-TO-SPARSE VIDEO CONTROL

We next study video control distillation in the dense-to-sparse setting using the Wan-VACE (Jiang et al., 2025) video generation model. The teacher receives a dense control sequence $\mathbf{p}_{\text{dense}}$, while the student observes only a sparse subset $\mathbf{p}_{\text{sparse}}$. Their conditions are

$$\begin{aligned} \mathbf{c}_T^+ &= (\mathbf{y}, \mathbf{p}_{\text{dense}}), & \mathbf{c}_S^+ &= (\mathbf{y}, \mathbf{p}_{\text{sparse}}), \\ \mathbf{c}_T^- &= (\emptyset, \mathbf{p}_{\text{dense}}), & \mathbf{c}_S^- &= (\emptyset, \mathbf{p}_{\text{sparse}}). \end{aligned} \quad (15)$$

The dense and sparse controls therefore induce different negative predictions even when both branches use the same null-text condition.

Table 1: Main comparison under joint multi-modal training. Each model is jointly trained on pose, depth, and scribble controls and evaluated separately on the three modalities. **Bold** and underlined numbers denote the best and second-best results, excluding Teacher.

Pose-Control		MPJPE ↓		PCK@0.2 ↑		PCK@0.1 ↑		Quality ↓	
Method		All	Key	All	Key	All	Key	FID	FVD
Teacher		3.03	2.82	96.02	96.46	91.15	92.23	14.13	56.01
Student		5.92	4.10	86.78	91.89	69.97	82.53	15.81	81.41
SFT		5.02	3.31	90.36	94.77	76.72	88.55	14.11	63.90
Off-policy		5.14	3.44	89.78	94.09	75.46	87.48	14.49	68.50
OPD (Naive)		4.43	3.14	91.84	95.25	80.28	89.35	<u>13.49</u>	<u>65.12</u>
OPD (IBM)		<u>4.20</u>	<u>2.88</u>	<u>92.93</u>	<u>96.11</u>	<u>82.27</u>	<u>90.95</u>	13.26	65.72
OPD (PDM)		4.13	2.83	92.98	96.12	82.48	91.03	13.52	62.77
Depth-Control		CORR ↑		RMSE ↓		SI-RMSE ↓		Quality ↓	
Method		All	Key	All	Key	All	Key	FID	FVD
Teacher		96.98	97.58	7.50	6.36	6.19	5.23	11.02	36.89
Student		83.21	91.27	16.87	11.13	14.36	9.37	15.03	71.33
SFT		88.39	96.17	13.83	7.80	11.96	6.52	12.87	58.13
Off-policy		88.25	96.01	13.98	8.00	12.10	6.64	12.66	<u>59.35</u>
OPD (Naive)		90.58	96.56	12.39	7.47	10.69	6.16	12.08	72.20
OPD (IBM)		<u>91.14</u>	<u>96.91</u>	<u>12.03</u>	<u>7.11</u>	<u>10.37</u>	<u>5.85</u>	<u>11.76</u>	60.38
OPD (PDM)		91.24	97.13	11.95	6.91	10.31	5.68	11.64	60.60
Scribble-Control		Recall ↑		Precision ↑		F1 ↑		Quality ↓	
Method		All	Key	All	Key	All	Key	FID	FVD
Teacher		93.26	94.40	94.26	95.27	93.73	94.81	9.56	24.22
Student		64.20	82.93	65.50	83.78	64.70	83.24	12.29	69.13
SFT		67.23	89.03	69.52	90.66	68.27	89.79	10.69	50.56
Off-policy		70.98	91.55	72.15	91.73	71.49	91.60	11.60	52.90
OPD (Naive)		74.78	90.77	75.38	91.34	75.00	90.99	10.53	59.95
OPD (IBM)		<u>75.92</u>	<u>92.23</u>	76.47	<u>92.70</u>	<u>76.13</u>	<u>92.43</u>	10.12	<u>52.67</u>
OPD (PDM)		76.42	93.03	<u>76.13</u>	92.86	76.22	93.03	<u>10.34</u>	50.53

5 EXPERIMENTS

We evaluate whether naive CFG-based OPD suffers from instability under NBA, and whether branch-aware objectives improve both optimization stability and generalization across guidance and conditioning variations.

5.1 EXPERIMENTAL SETUP

Visual knowledge distillation. We evaluate the visual knowledge distillation setting introduced in Section 4.3.1 on held-out prompts spanning rendering style and concept identity. We use this experiment as a qualitative case study of NBA: all students are trained at a fixed guidance scale ($\gamma_{\text{train}} = 3$) and evaluated across different inference guidance scales, revealing whether guided-only matching recovers transferable branch-level predictions or merely fits a particular CFG composition. We compare naive CFG-based OPD with PDM under the same training setup. Additional details are provided in Appendix 7.1.1.

Dense-to-sparse video control. We construct a dense-to-sparse video control benchmark from OpenHumanVideo (Li et al., 2025), containing 600 test clips across diverse subjects, motions, and interaction scenarios. We evaluate pose, depth, and scribble control under the dense-to-4-keyframe setting, with all methods evaluated on full video rollouts against the ground-truth videos.

Table 2: Guidance-scale generalization in the pose-only ablation setting. All models are trained only on pose control at a fixed guidance scale $\gamma_{\text{train}} = 5$ and evaluated at different inference guidance scales.

Method	MPJPE ↓		PCK@0.2 ↑		PCK@0.1 ↑		Quality ↓	
	All	Key	All	Key	All	Key	FID	FVD
Naive ($\gamma = 5$)	4.62	3.22	91.35	95.01	79.39	89.54	14.05	67.44
Naive ($\gamma = 3$)	4.94	3.31	90.23	94.53	77.92	88.22	22.66	134.09
Naive ($\gamma = 1$)	8.98	6.23	80.08	87.42	69.14	81.57	78.20	507.70
IBM ($\gamma = 5$)	4.28	2.91	92.67	95.95	81.69	91.03	12.99	53.83
IBM ($\gamma = 3$)	4.42	3.04	92.09	95.53	81.41	90.81	13.69	56.85
IBM ($\gamma = 1$)	4.91	3.43	90.92	94.69	80.66	89.98	18.00	71.17
PDM ($\gamma = 5$)	4.27	2.75	92.71	96.38	82.12	91.66	13.83	58.65
PDM ($\gamma = 3$)	4.17	2.73	92.81	96.61	82.27	91.81	13.46	55.85
PDM ($\gamma = 1$)	4.48	2.90	92.14	95.64	81.72	91.15	15.25	60.97

Unless otherwise specified, we use $\lambda = 1$ for PDM and supervise the first $K = 8$ student-visited transitions. For the main comparison, each method is jointly trained on pose, depth, and scribble controls and evaluated separately on each modality. The guidance-scale and hyperparameter ablations instead use pose-only models to isolate the factor being studied and reduce repeated training cost. Results obtained under the joint-modality and pose-only protocols are therefore not directly comparable. Complete benchmark, evaluation, and implementation details are provided in Appendix 7.1.

5.2 MAIN RESULTS

5.2.1 VISUAL KNOWLEDGE DISTILLATION

As shown in Figure 2, naive CFG-based OPD struggles to transfer reference-supplied visual knowledge consistently, despite directly matching the final guided prediction. In contrast, branch-aware objectives more effectively preserve the defining visual attributes provided by the reference image, demonstrating the importance of branch-level supervision under asymmetric conditioning.

5.2.2 DENSE-TO-SPARSE VIDEO CONTROL

Dense-to-Sparse Distillation under Training Guidance We first evaluate all methods using joint training over pose, depth, and scribble controls. Table 1 reports results under the default guidance configuration, where the training and inference guidance scales are both set to $\gamma = 5$. Naive CFG-based OPD consistently underperforms branch-aware objectives across different modalities, despite using the same CFG configuration during training and inference. This indicates that matching only the composed CFG prediction provides insufficient supervision even when the guidance scale remains unchanged.

Empirically, PDM delivers the most consistent overall improvements across control modalities, while IBM also outperforms naive guided matching on most control-fidelity metrics.

Generalization across Guidance Scales The previous experiments evaluate all methods under the same guidance configuration used during training. However, NBA predicts that matching only the CFG-composed prediction does not identify the underlying positive and negative branches, and therefore may fail when the guidance composition changes.

Table 2 evaluates the generalization of different objectives across inference guidance scales. Naive OPD exhibits severe degradation when the guidance scale deviates from the training configuration, with the performance dropping substantially at $\gamma = 1$. In contrast, branch-aware objectives remain robust across different guidance scales, confirming that explicitly constraining branch-level information provides more transferable supervision.

Both branch-aware objectives substantially improve guidance-scale robustness over naive OPD. In this pose-control setting, PDM exhibits the most stable performance across inference guidance scales.



Figure 3: Qualitative comparison on dense-to-sparse pose control distillation under the default guidance scale $\gamma = 5$. Compared with existing distillation objectives, Positive-Direction Matching (PDM) produces more faithful motion trajectories and higher visual quality under sparse control supervision.

Table 3: Effect of the conditional-direction weight λ in the pose-only ablation setting. **Bold** and underlined numbers denote the best and second-best results, respectively.

Method	MPJPE ↓		PCK@0.2 ↑		PCK@0.1 ↑		Quality ↓	
	All	Key	All	Key	All	Key	FID	FVD
PDM ($\lambda = 0$)	4.24	<u>2.81</u>	92.71	96.24	<u>81.80</u>	<u>91.05</u>	13.97	59.72
PDM ($\lambda = 1$)	<u>4.27</u>	2.75	92.71	96.38	82.12	91.66	13.83	58.65
PDM ($\lambda = 3$)	4.36	2.95	<u>92.08</u>	95.77	81.25	90.39	13.33	61.79
PDM ($\lambda = 5$)	4.40	3.00	<u>92.27</u>	95.96	80.89	90.75	<u>12.97</u>	56.85
PDM ($\lambda = 10$)	4.49	3.07	91.60	95.35	80.21	89.93	12.96	<u>56.91</u>

5.3 ABLATION STUDIES

5.3.1 EFFECT OF CONDITIONAL-DIRECTION WEIGHT

We study the effect of λ in PDM, which controls the strength of conditional-direction preservation. When $\lambda = 0$, PDM reduces to positive-only matching and does not explicitly constrain the CFG conditional direction.

As shown in Table 3, $\lambda = 1$ provides the best overall balance between control fidelity and generation quality. In particular, it achieves the lowest keyframe MPJPE and the highest PCK@0.1 scores, while also improving FVD over positive-only matching. Increasing λ beyond 1 improves some distributional-quality metrics, with $\lambda = 5$ and $\lambda = 10$ producing lower FID and FVD, but consistently weakens control accuracy. This suggests that overemphasizing direction matching introduces a trade-off between visual quality and adherence to the sparse control signal.

Table 4: Effect of the OPD supervision horizon K in the pose-only ablation setting. Wall-clock training time is reported in seconds per optimization step. **Bold** and underlined numbers denote the best and second-best results, respectively.

Method	MPJPE ↓		PCK@0.2 ↑		PCK@0.1 ↑		Quality ↓		Time ↓
	All	Key	All	Key	All	Key	FID	FVD	s/step
PDM ($K = 1$)	4.33	2.98	92.53	95.68	81.19	90.57	<u>13.70</u>	54.41	~23
PDM ($K = 2$)	4.45	3.11	92.18	95.57	80.78	90.51	13.89	60.63	~38
PDM ($K = 4$)	4.27	<u>2.86</u>	<u>92.59</u>	<u>95.98</u>	<u>81.71</u>	<u>91.13</u>	13.58	<u>57.08</u>	~69
PDM ($K = 8$)	4.27	2.75	92.71	96.38	82.12	91.66	13.83	58.65	~132
PDM ($K = 50$)	4.65	3.14	91.34	95.25	80.81	90.09	13.91	61.95	~790

Because control fidelity is the primary objective in our dense-to-sparse setting, while generation quality must remain competitive, we use $\lambda = 1$ as the default value in all main experiments.

5.3.2 EFFECT OF ON-POLICY SUPERVISION HORIZON

We further study the number of student-visited denoising transitions used for OPD supervision. During training, the student generates a full denoising trajectory, while only the first K visited states contribute to the distillation objective.

Table 4 shows that increasing K substantially increases training cost, from 23 seconds per step at $K = 1$ to 790 seconds per step when supervising the full 50-step trajectory. Among the intermediate settings, $K = 8$ achieves the strongest control fidelity, including the lowest keyframe MPJPE and the highest PCK scores, while retaining substantially lower cost than full-trajectory supervision. Although $K = 1$ and $K = 4$ provide better individual distributional-quality metrics and lower training cost, they are less accurate in following the sparse control signals. We therefore use $K = 8$ as the default supervision horizon, prioritizing control fidelity while avoiding the prohibitive cost and degraded performance of full-trajectory supervision.

5.4 QUALITATIVE RESULTS

Figure 2 and Figure 3 present qualitative comparisons for visual knowledge distillation and dense-to-sparse video control respectively.

For more qualitative results, please refer to Appendix 7.2.

6 CONCLUSIONS AND LIMITATIONS

We identify Negative Branch Asymmetry (NBA), a failure mode of CFG-based OPD in which teacher–student discrepancies on the positive and negative branches can cancel under guided prediction matching. To address this ambiguity, we introduce Positive–Direction Matching (PDM), which supervises the positive prediction and CFG conditional direction separately. Experiments on visual knowledge distillation and dense-to-sparse video control show that PDM improves distillation performance and remains more stable across inference guidance scales.

Our study is limited to linear CFG and two model families, while the visual knowledge results are primarily qualitative. Moreover, PDM and IBM share the same branch-level zero-loss solution under non-degenerate weights, so PDM’s advantage is currently empirical rather than theoretical. Future work may extend NBA analysis to other guidance mechanisms and develop more efficient on-policy supervision.

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Bucket	Motion score	#Test	#Train
diff1.5-2.0	[1.5, 2.0)	100	2,194
diff2.0-2.5	[2.0, 2.5)	100	6,355
diff2.5-3.5	[2.5, 3.5)	100	8,992
diff3.5-5.0	[3.5, 5.0)	100	4,851
diff5.0-7.0	[5.0, 7.0)	100	1,319
diff7.0+	[7.0, ∞)	100	364
Total		600	$\approx 24\text{K}$

Table 5: Difficulty-stratified benchmark. The test set draws 100 clips from each of six motion-difficulty buckets (600 total); motion score is the mean frame-to-frame pixel difference of the rendered DWPose skeleton.

7 APPENDIX

7.1 ADDITIONAL EXPERIMENTAL DETAILS

7.1.1 VISUAL KNOWLEDGE DISTILLATION

bn: Haozhe’s part, TBD

7.1.2 DENSE-TO-SPARSE VIDEO BENCHMARK

Our dense-to-sparse video benchmark is derived from OpenHumanVid (OHV) (Li et al., 2025). Starting from the 12-part OHV release ($\approx 91\text{K}$ clips with DWPose annotations), we first filter clips according to minimum length and motion magnitude. We then split videos into non-overlapping 81-frame clips at 832×480 resolution while preserving pixel alignment between RGB and control streams.

Difficulty stratification. To reduce evaluation bias toward easier samples (Li et al., 2026a), we compute a motion score based on frame-to-frame differences of rendered DWPose skeletons. The clips are divided into six difficulty buckets according to this score, and we randomly sample 100 clips from each bucket to form the 600-clip test set. The remaining clips are used for training and validation. Please refer to Table 5 for more details.

7.1.3 SPARSE KEYFRAME CONTROL

All video experiments target the dense-to-sparse setting: the teacher receives the control signal at *every* frame, while the student receives it only at a sparse set of pixel-space keyframes. Following the masked video-to-video (MV2V) interface of the VACE (Jiang et al., 2025) base model, the student input frame stack F and mask M (over 81 frames at 480×832 , in the $[-1, 1]$ range) are built as

- $F[0]$ = the reference image (appearance anchor), with $M[0] = 0$ (given, not generated);
- $F[k]$ = the control frame at each pose keyframe $k \in \{20, 40, 60\}$;
- all remaining frames are blank (-1), with $M[\cdot] = 1$ (to be synthesized).

We use the uniform layout `pixel_kf` = $\{0, 20, 40, 60\}$ throughout: index 0 is reserved for the reference image, so the model is conditioned on pose at three interior keyframes out of 81 frames. Keyframes are indexed in *pixel* space; under the VACE VAE ($4 \times$ temporal, $8 \times$ spatial compression) this maps non-linearly to latent frames, so all masks are constructed in pixel space and then compressed. This same sparse construction is used both inside the on-policy rollout during training and at inference; the teacher instead receives the dense per-frame control.

7.1.4 EVALUATION METRICS

All methods are evaluated independently on the 600-clip test set. Scores are computed per clip and averaged over the dataset.

<i>Base model & adapter</i>	
Base model	Wan2.1-VACE-1.3B (frozen)
Adapter	LoRA, rank 32, $\alpha = 32$
LoRA target	self/cross-attn $\{q, k, v, o\}$, ffn $\{0, 2\}$ per VACE block
Precision	bf16 autocast, gradient checkpointing
<i>Optimization</i>	
Optimizer	AdamW, $\beta = (0.9, 0.999)$, wd = 0, $\epsilon = 10^{-8}$
Learning rate	1×10^{-4} (constant)
Training steps	2000
Devices / batch	8 GPUs (DDP), per-GPU batch 1
<i>Diffusion & rollout</i>	
Solver / schedule	UniPC, 50 steps, <code>shift</code> = 16
OPD rollout K	8 active steps ("8-of-50")
Guidance scale γ	5.0
<i>Data</i>	
Resolution / length	480×832 , 81 frames
Keyframes	pixel $\{0, 20, 40, 60\}$ (0 = reference)
Train / val / test	$\approx 24\text{K} / 24 / 600$ clips

Table 6: Training and inference hyperparameters, shared across all distillation objectives.

Control fidelity (per modality). For each control modality we *re-extract* the control signal from the generated video with the corresponding annotator and compare it to the control that was fed in.

- **Pose.** We run DWPose (Yang et al., 2023) on the generated video and use the 18 body joints, in normalized $[0, 1]$ image coordinates. A joint is counted only when both the GT and predicted confidence exceed 0.3; frames with fewer than 3 valid joints are skipped. We report *MPJPE* (mean per-joint ℓ_2 error in normalized coordinates) and *PCK@ τ* for $\tau \in \{0.2, 0.1\}$, where a joint is correct if its error is below $\tau \cdot s$ and s is the per-frame bounding-box size $s = \max(\Delta x, \Delta y)$ of the GT skeleton. Metrics are reported over *all* frames and over the *keyframe* subset separately.
- **Depth.** We re-extract depth from the generated frames with DPT-Hybrid (MiDaS) (Ranftl et al., 2020) and compare against the dense depth control (same annotator). Since these are *relative* depths, we remove the scale/shift ambiguity with a closed-form least-squares affine alignment $a \cdot \hat{d} + b \approx d$ and report the affine-invariant *SI-RMSE* (primary), the raw RMSE, and the Pearson correlation.
- **Scribble.** We re-extract neural line-art from the generated frames with the VACE scribble annotator and compare against the dense line-art control. Line pixels are binarized as deviations from the per-frame median ($> 40/255$), and we report a tolerant boundary *F1* (precision/recall) with a dilation tolerance of 2 pixels.

Distributional quality. Independent of control adherence, we measure perceptual and temporal realism against the GT distribution: *FID* (clean-fid, InceptionV3 features, 8 frames per clip) captures per-frame appearance, and *FVD* (I3D pretrained on Kinetics-400, 16 frames linspace-sampled per clip) captures spatio-temporal consistency.

7.1.5 IMPLEMENTATION AND HYPERPARAMETERS

All methods fine-tune the same Wan2.1-VACE-1.3B (Jiang et al., 2025) base model with LoRA (Hu et al., 2022) adapters (the base weights are frozen), so differences are attributable to the training objective alone. LoRA (rank 32, $\alpha = 32$) is applied to the query/key/value/output projections of both self- and cross-attention and the two feed-forward linears within every VACE block. We train with AdamW ($\text{lr} = 10^{-4}$, $\beta = (0.9, 0.999)$, no weight decay) for 2000 steps on 8 GPUs (DDP, per-GPU batch 1), under bf16 mixed precision with gradient checkpointing. Inference and the on-policy rollout use the UniPC solver (Zhao et al., 2023) on a 50-step schedule (`shift`= 16). All models are distilled and deployed at classifier-free guidance scale $\gamma = 5$. Videos are 81 frames at 480×832 . Table 6 lists the full configuration.

7.2 MORE QUALITATIVE RESULTS

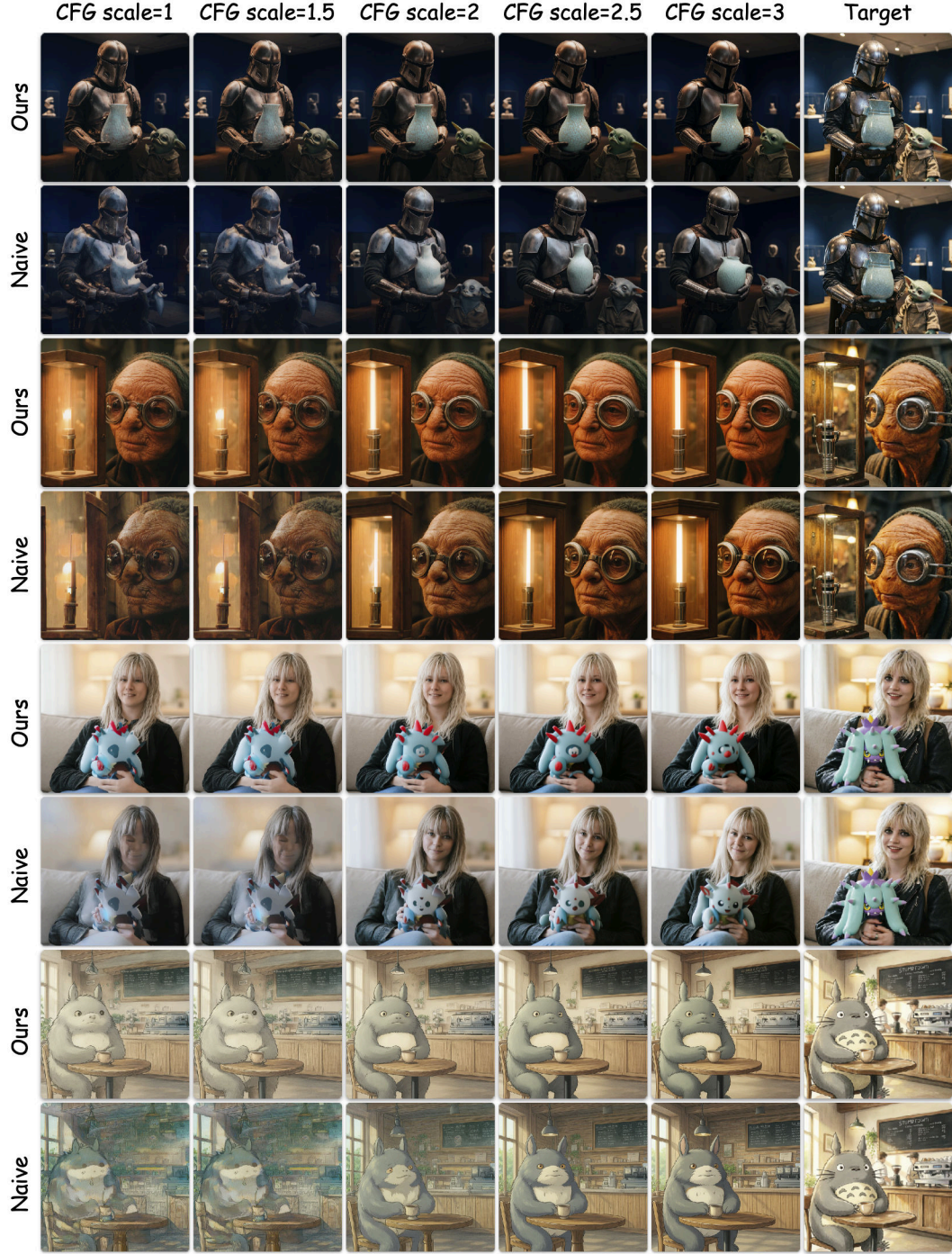


Figure 4: Qualitative comparison on concept identity distillation across inference CFG scales. The rightmost column shows the target appearance to be internalized by the text-only student. PDM (Ours) consistently preserves the defining visual attributes of each target across guidance scales, whereas naive OPD exhibits identity drift and degraded generation quality, particularly when the inference guidance scale changes. These results illustrate the guidance-scale sensitivity induced by NBA.



Figure 5: Qualitative comparison on dense-to-sparse depth control distillation under the default guidance scale $\gamma = 5$. The teacher receives dense depth guidance, while the student is conditioned only on sparse keyframes. PDM achieves more consistent structure and visual quality compared with other distillation objectives.



Figure 6: Qualitative results on dense-to-sparse scribble control distillation with $\gamma = 5$. Compared with other distillation objectives, Positive-Direction Matching better preserves the scribble-guided structure and generates more consistent visual results under sparse control supervision.